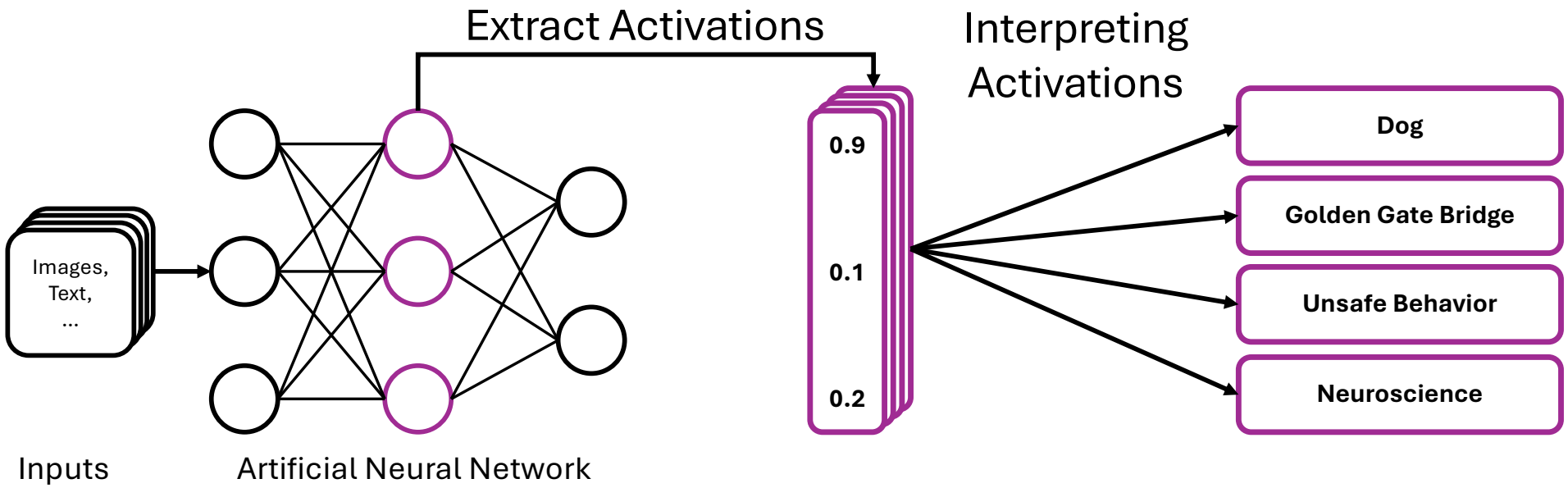
Backward Factor Traces (BFT)

Reinterpreting Interpretability

What is Mechanistic Interpretability?



- Aim to understand internal mechanisms of computations
- Operates on the space of activations

Limitations

- Interpretability on mechanisms tells us which neurons are active
  - **Not** why they are active
- This makes it difficult to causally trace activity through the network
- Do the activation patterns of a single layer tell us anything about actual circuits or mechanisms implemented by the network?

Understanding Mechanisms in Neuroscience

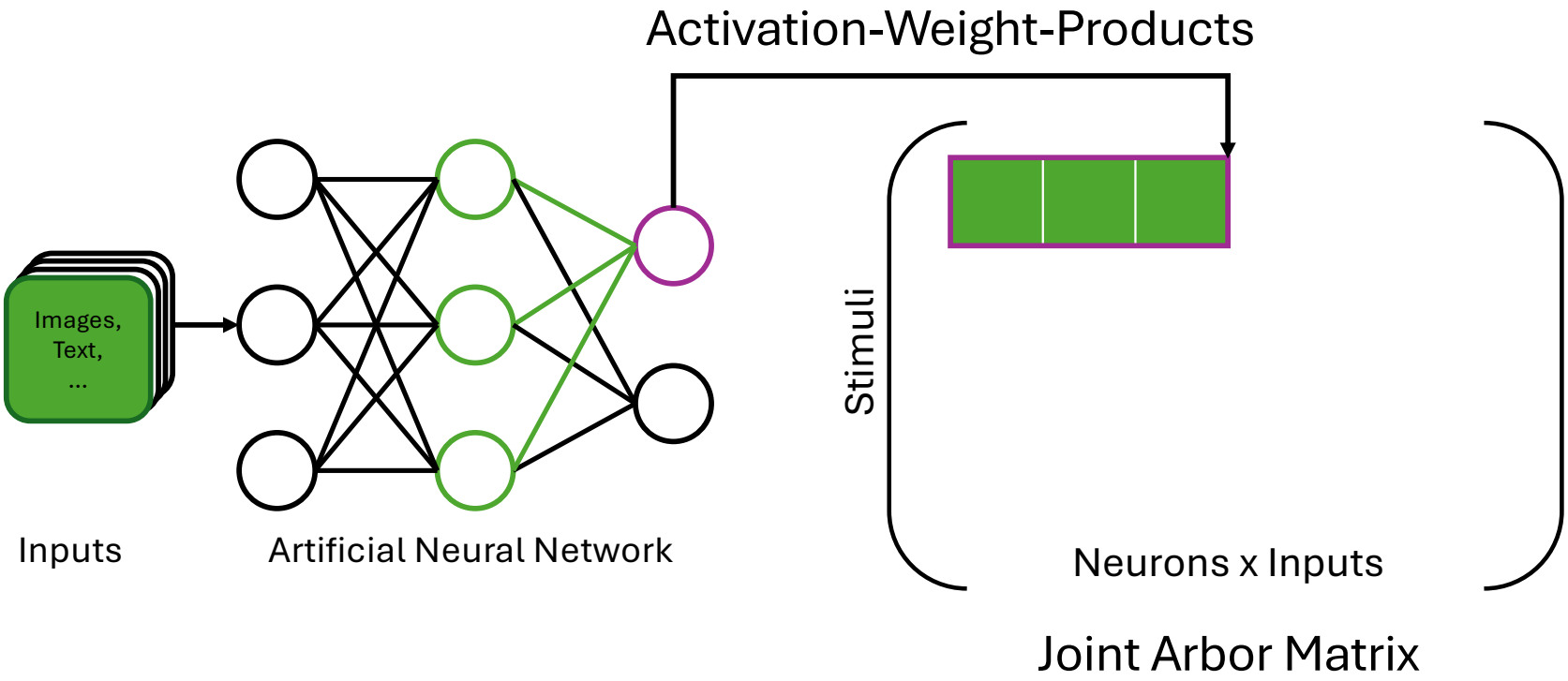
- Circuit level:
  - Which neurons are connected?
  - What task does a circuit solve?

- Dendritic arbors:
  - Ask what lead to the neuron being active
  - Arbors themselves already are complex networks

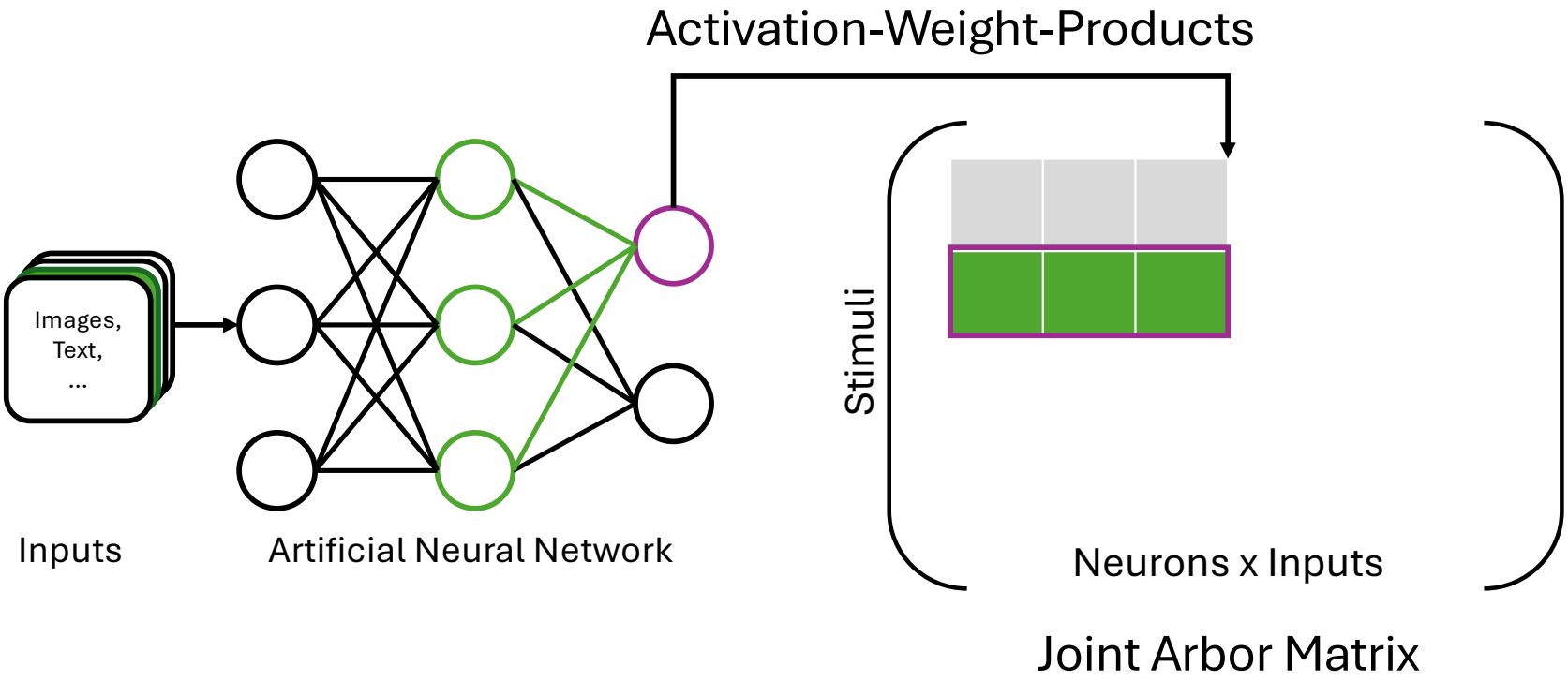
→ Apply these ideas to interpretability

Method: Backward Factor Traces

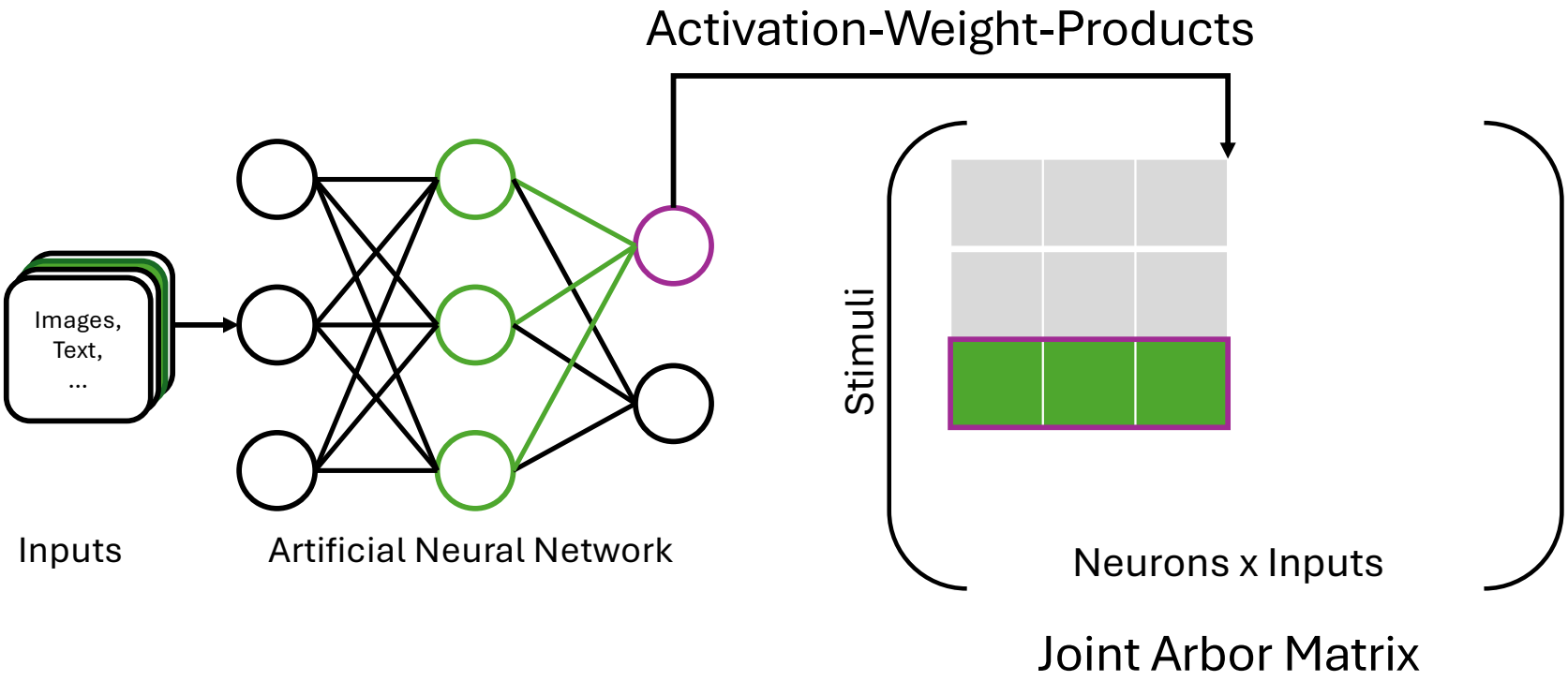
From Activations to Arbors



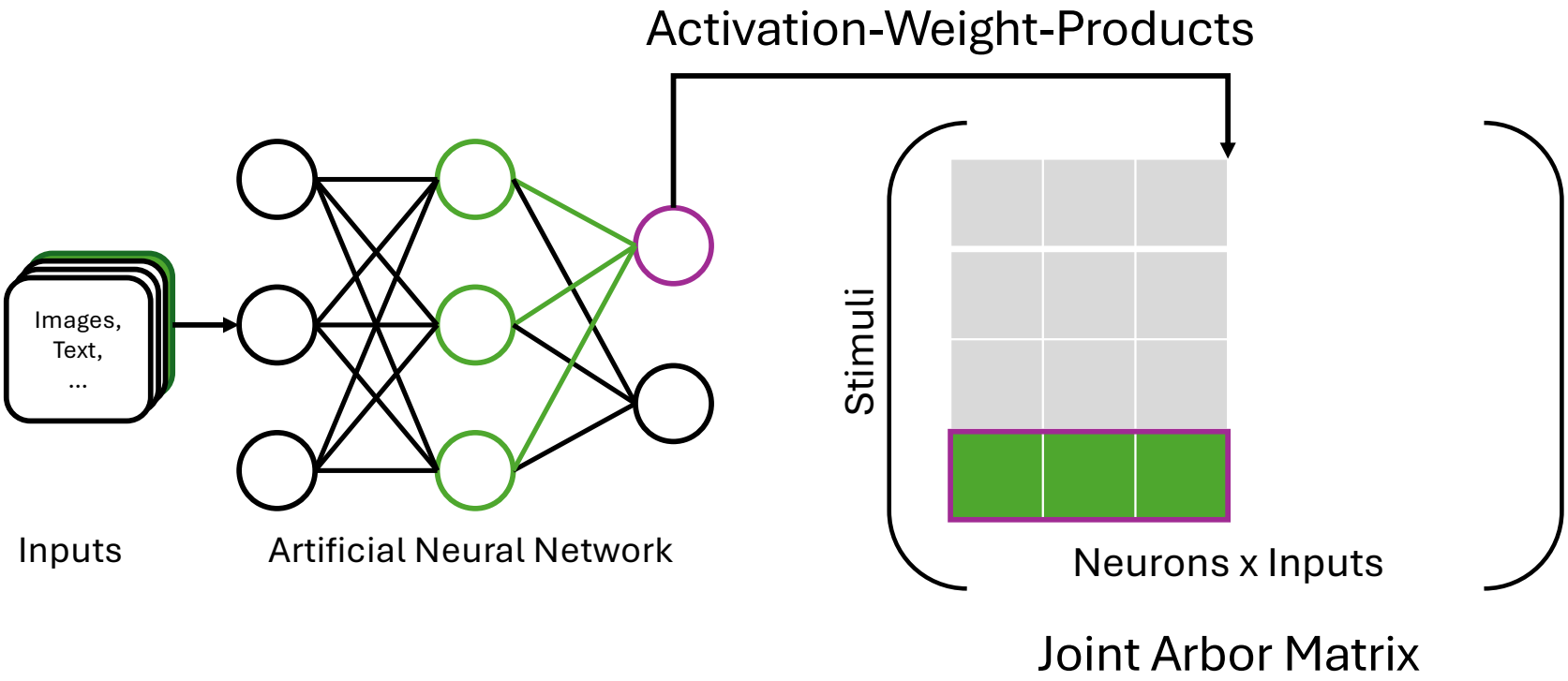
From Activations to Arbors



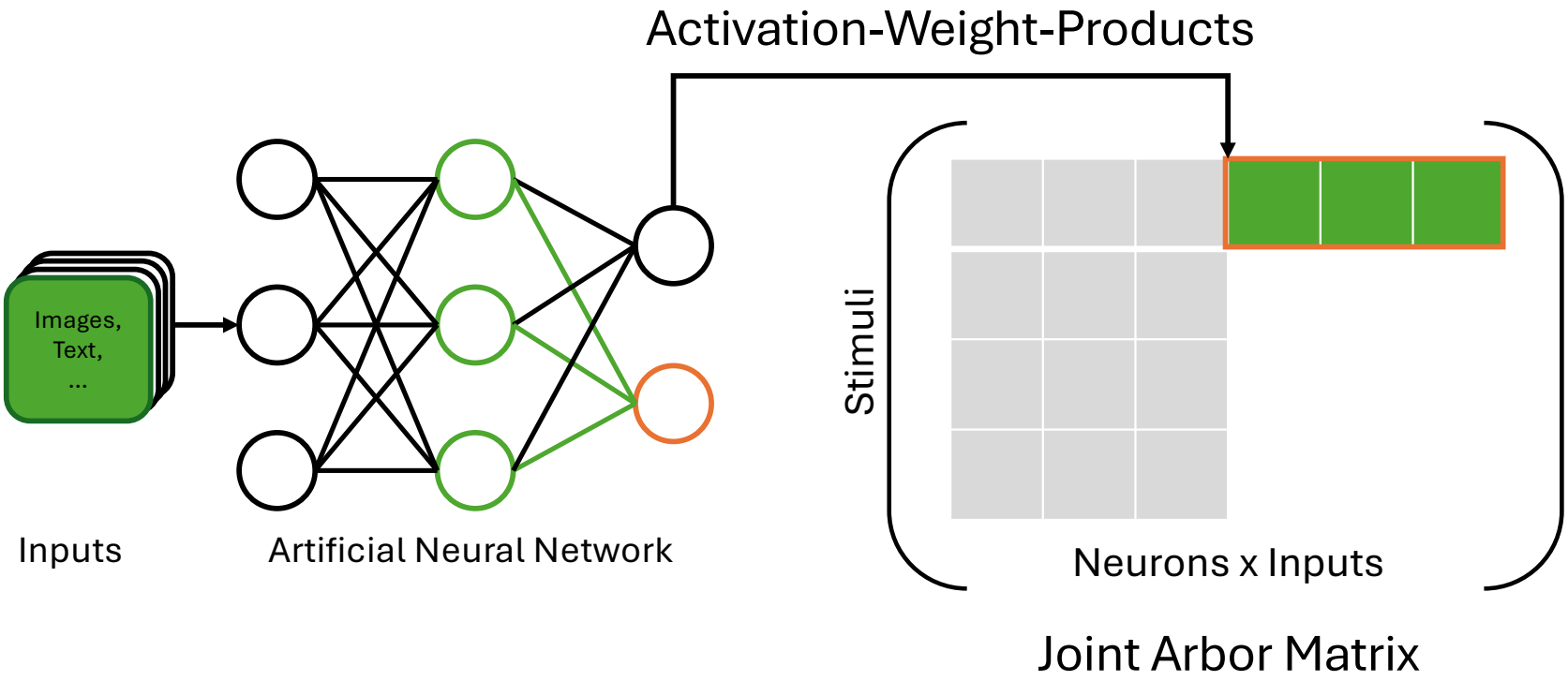
From Activations to Arbors



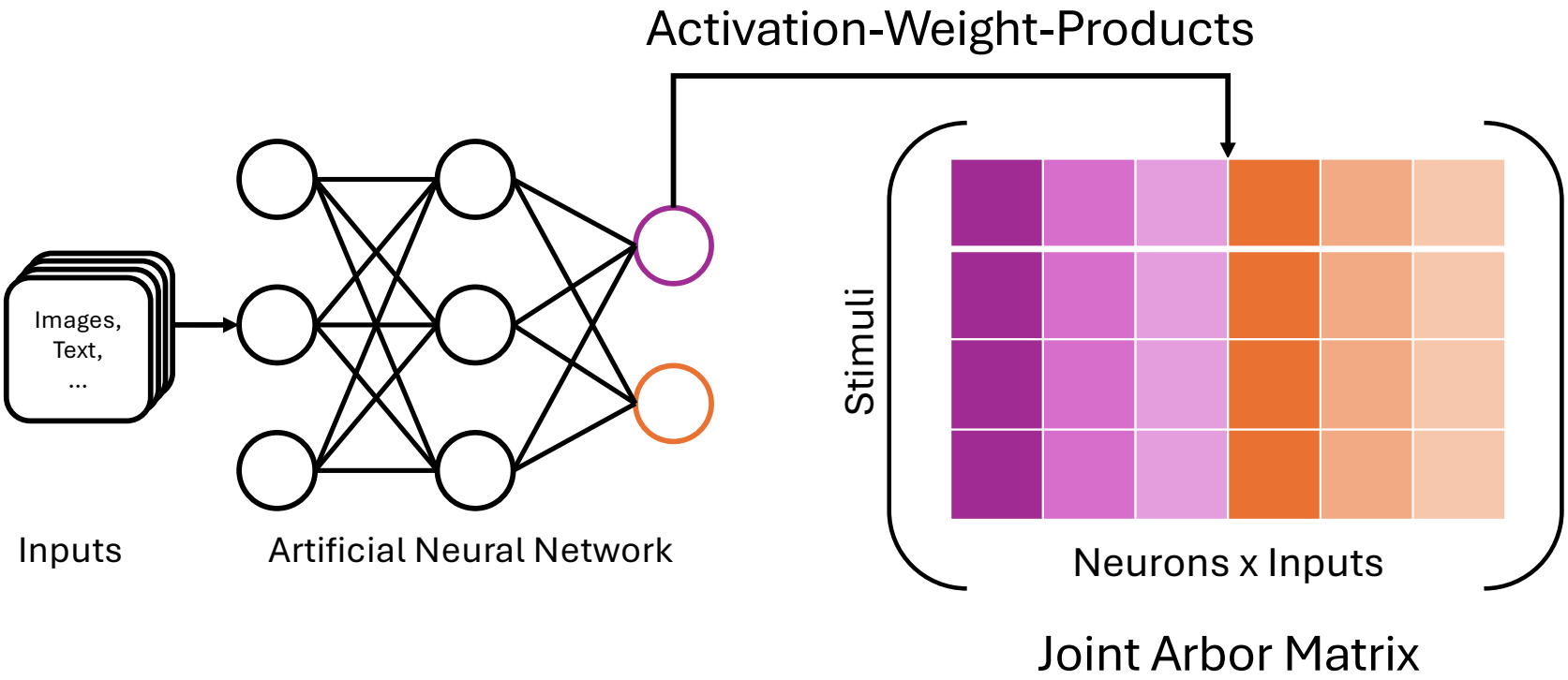
From Activations to Arbors



From Activations to Arbors



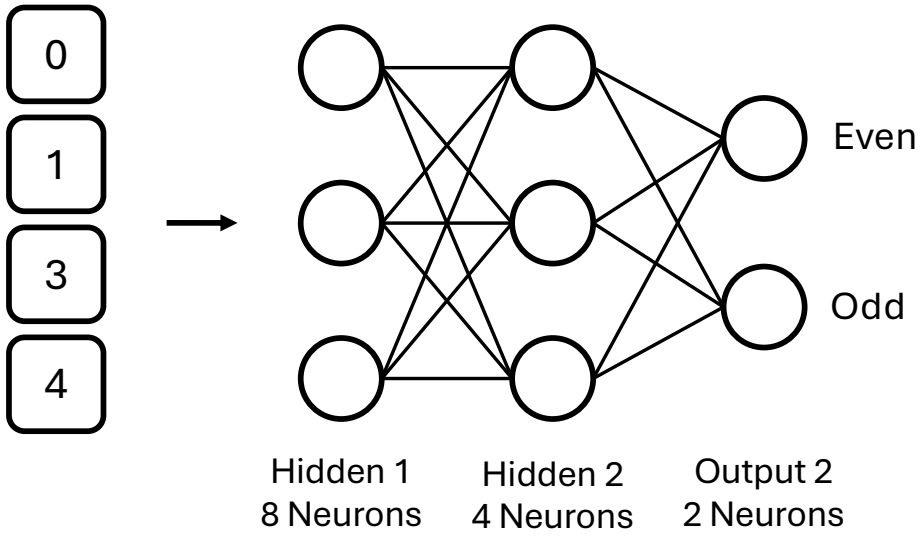
From Activations to Arbors



Toy Example: Even/Odd MNIST

- MNIST as very simple controlled dataset
- Use only digits 0, 1, 3, 4
  - Two classes Even vs Odd
  - Each with two subclasses that we know about

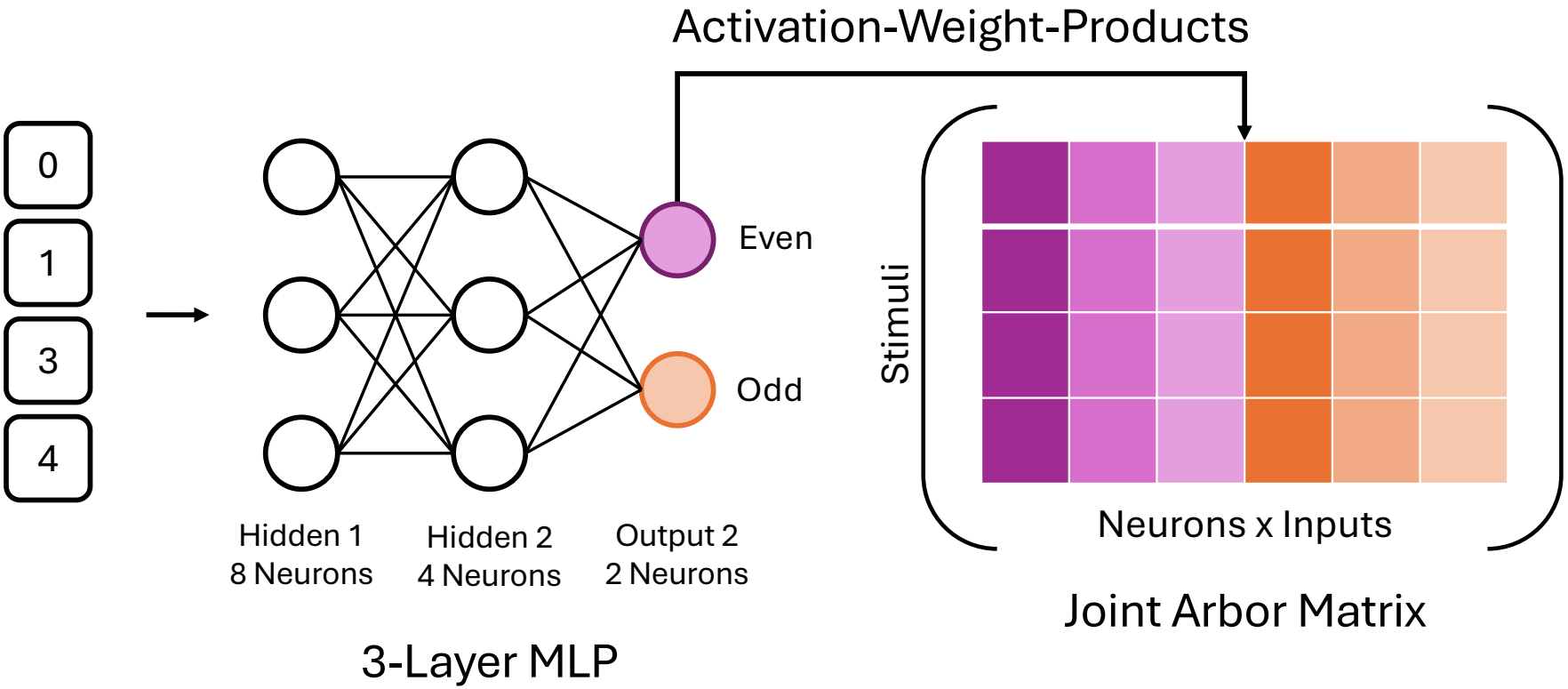
Toy Example: Even/Odd MNIST



3-Layer MLP

- Simple and controlled task
- Two classes
  - Each consisting of two sub-classes

Factorizing Arbors



Non-Negative Matrix Factorization (NMF)

$$A \approx \sum_{k=1}^{K} \lambda_k \, \mathbf{w}_k \otimes \mathbf{h}_k$$

Subject to:

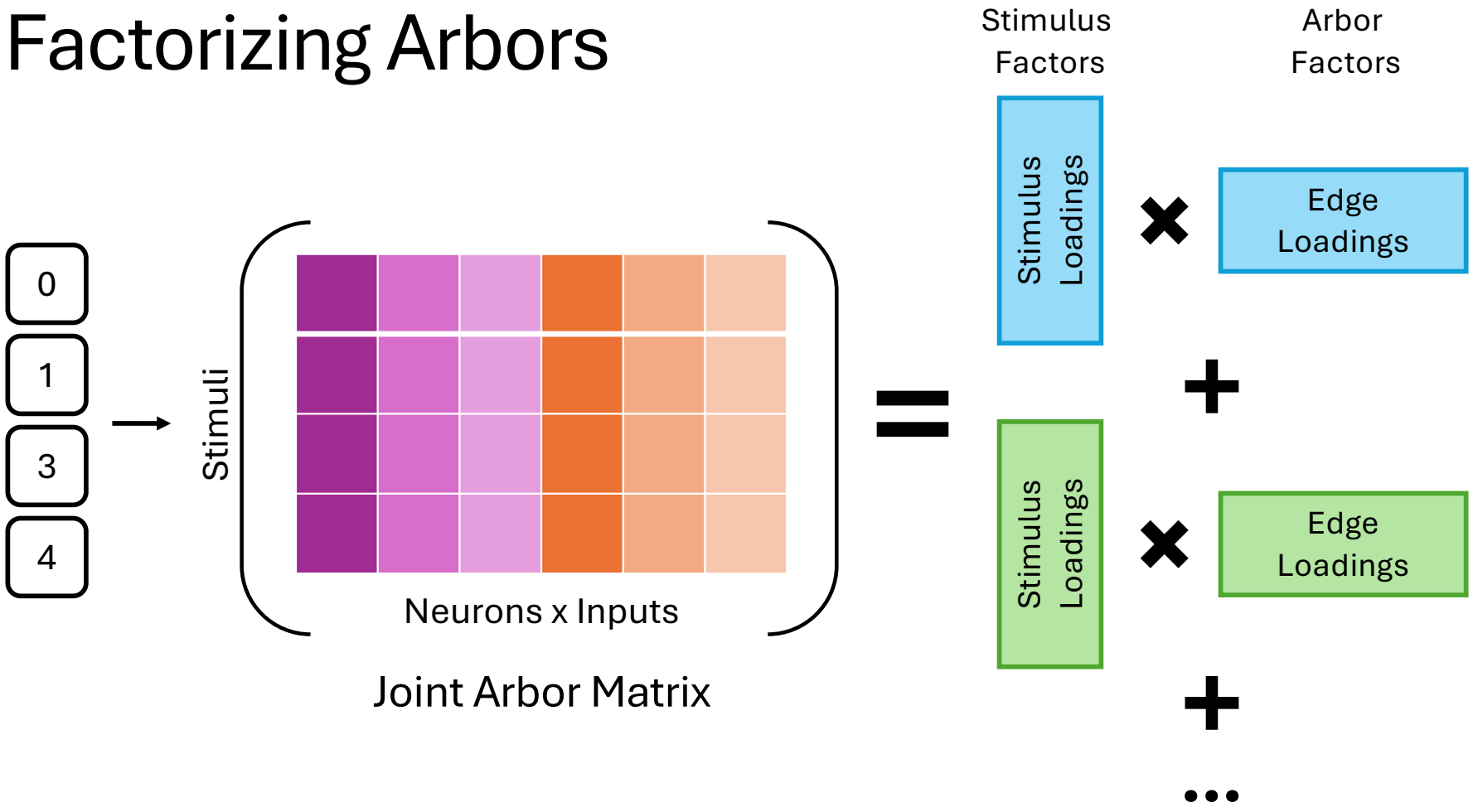
$$\mathbf{w}_k \in \mathbb{R}^m_{\geq 0}, \quad \mathbf{h}_k \in \mathbb{R}^n_{\geq 0},$$

$$\|\mathbf{w}_k\|_2 = \|\mathbf{h}_k\|_2 = 1, \quad \lambda_k \geq 0$$

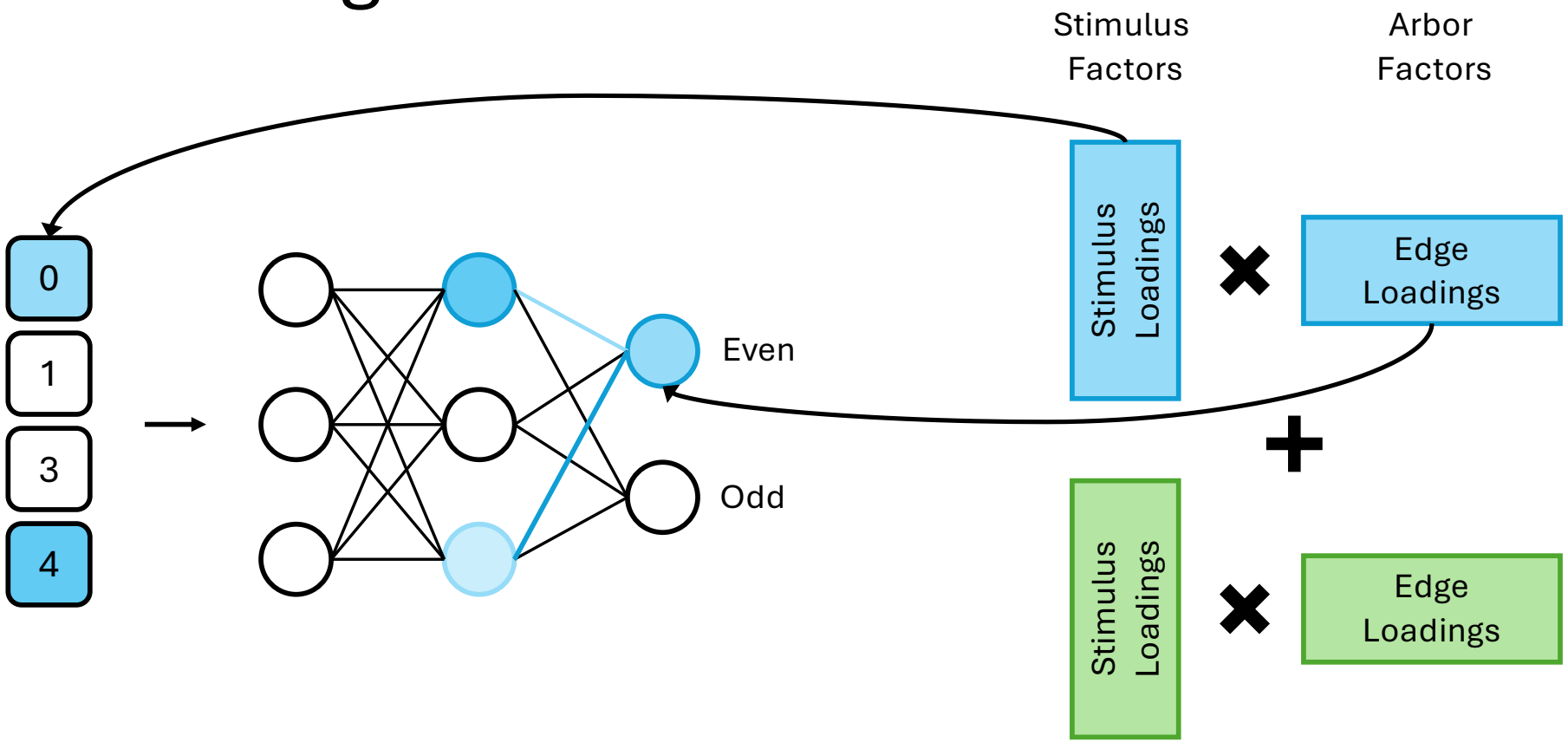
Optimize:

$$\min_{\lambda_k, \, \mathbf{w}_k, \, \mathbf{h}_k} \left\| A - \sum_{k=1}^{r} \lambda_k \, \mathbf{w}_k \mathbf{h}_k^\top \right\|_F^2$$

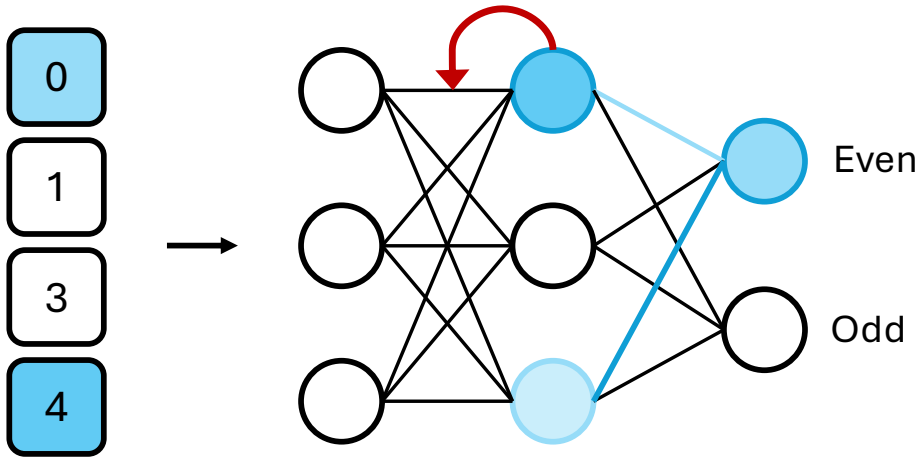
Factorizing Arbors



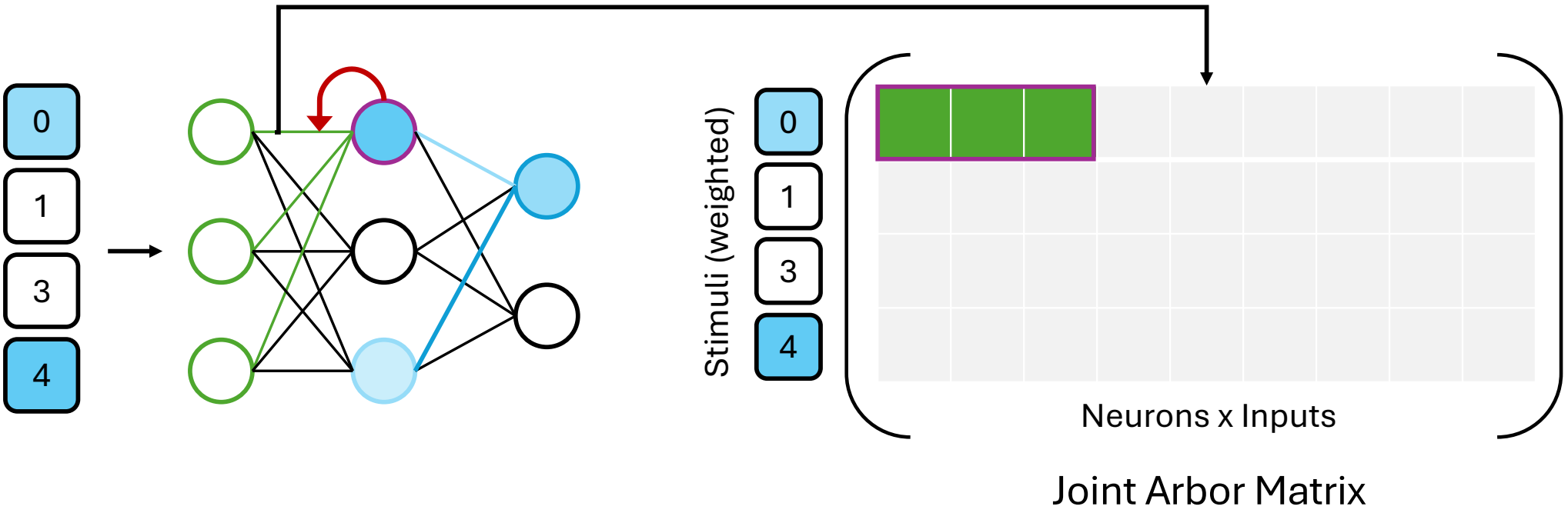
Factorizing Arbors



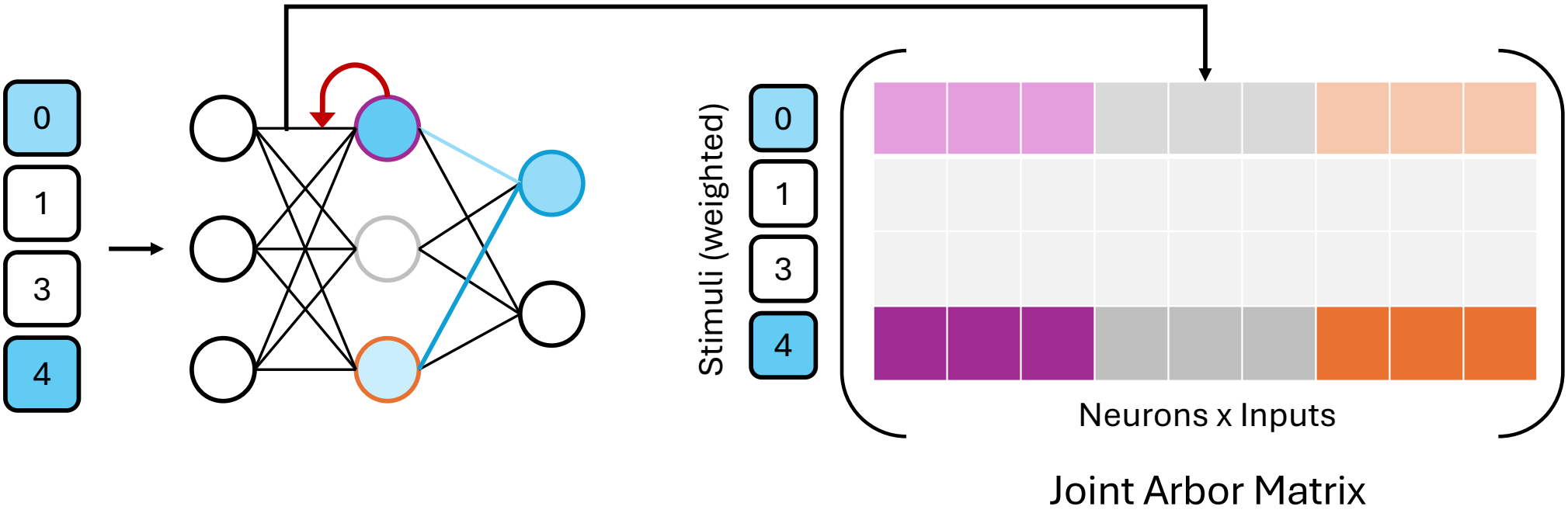
Tracing Factors Backward



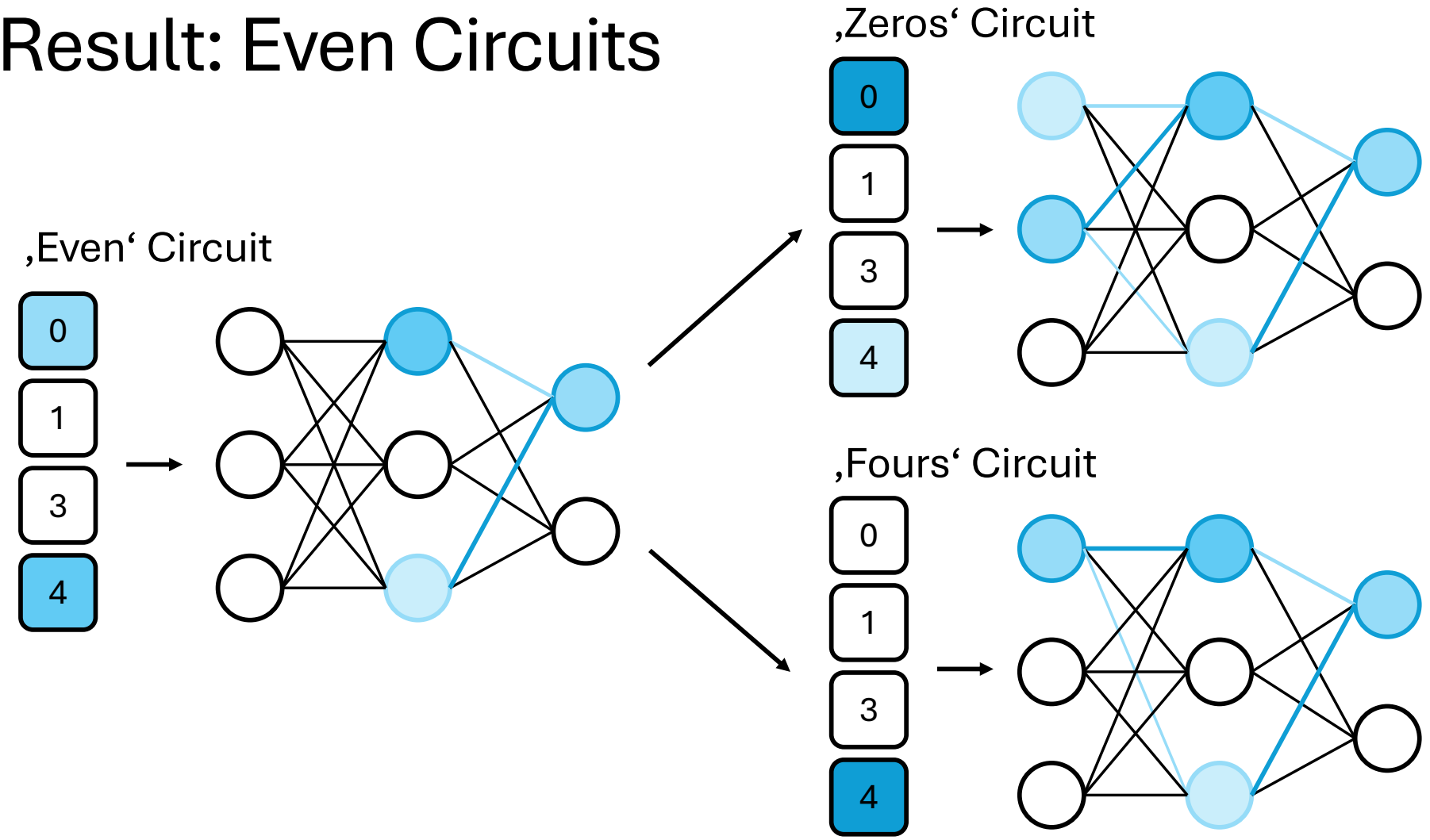
Tracing Factors Backward



Tracing Factors Backward

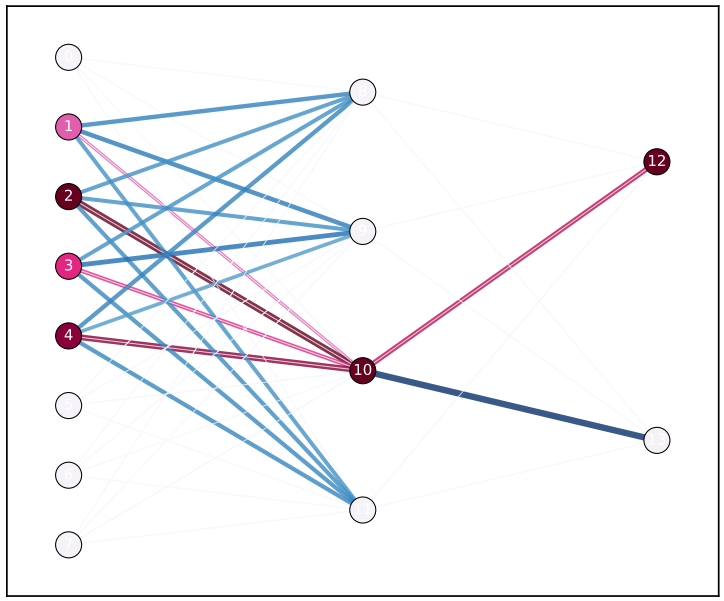


Result: Even Circuits



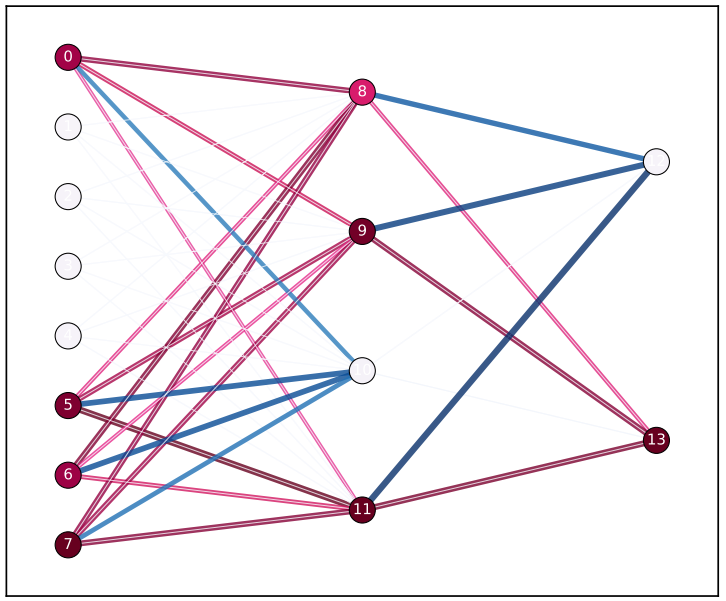
Even-Odd Results: Circuits

Scaffold Graph [path: F1→F0]



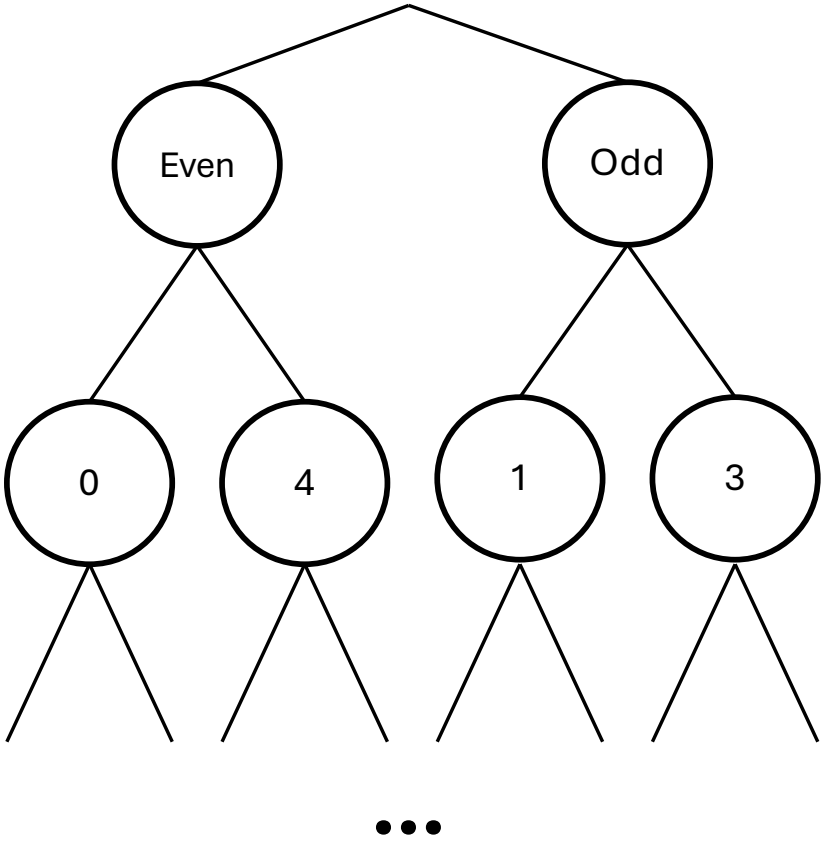
Even

Scaffold Graph [path: F0→F0]

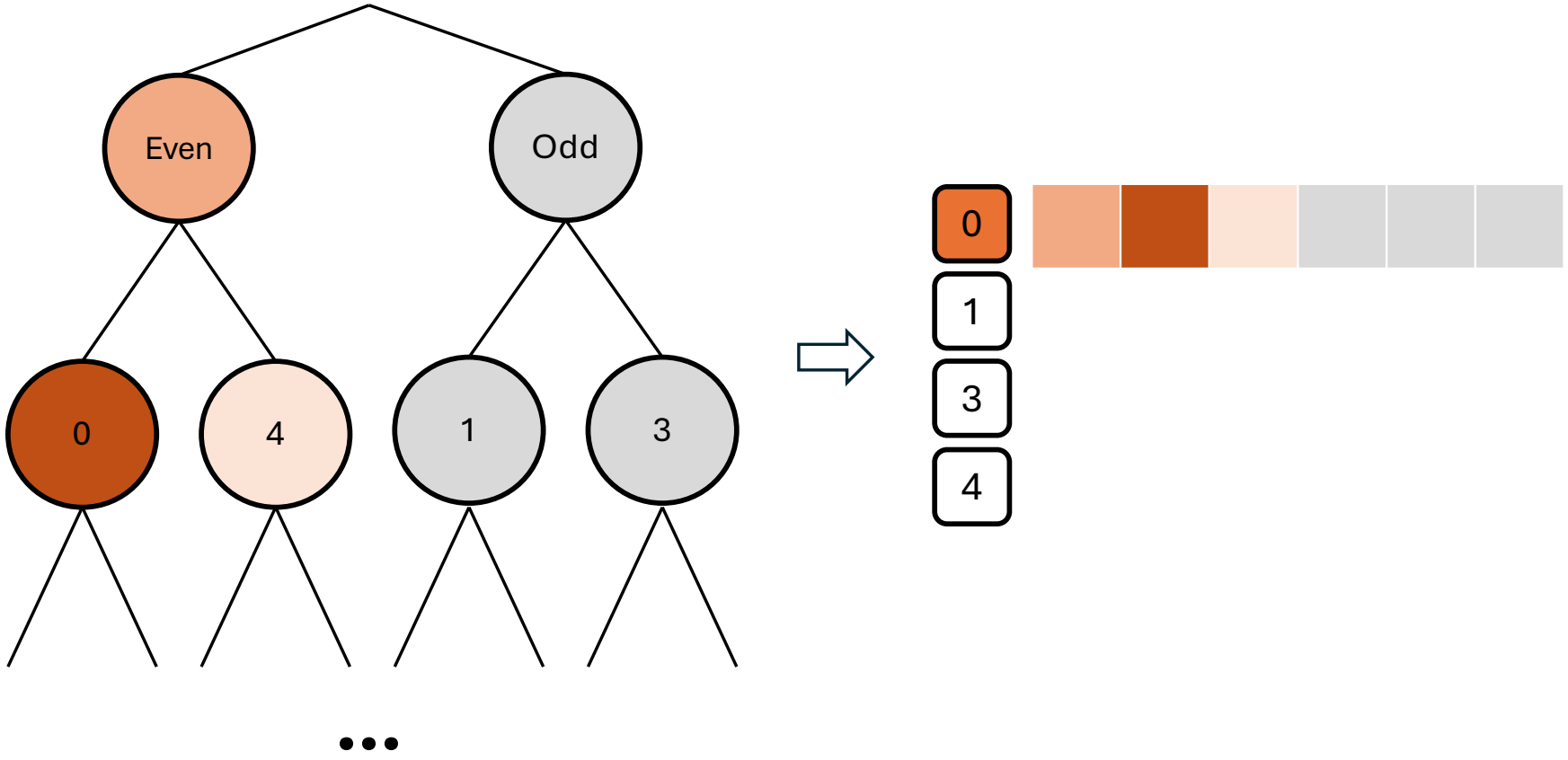


Odd

Factor Tree



Fingerprints



Fingerprints

